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High-Performance System Administration

Introduction to Network File System (NFS)

Learning Objectives

After the course the students should be able to:

- Describe common usage of NFS
- Deploy NFS infrastructure on a server and a client
- Examine NFS deployments

Outline

- 1 Whats NFS
- 2 Recap Terminal Editors
- 3 Installation of an NFS server
- 4 Installation of an NFS client

What is NFS

NFS (either Network File System or Network File Service) is the most common protocol for sharing files between Unix systems over a network. NFS servers export directories from their local hard disks to NFS clients, which mount them so that they can be accessed like any other directory.

How is NFS structured

- Server / client architecture
- The server
 - ▶ Shares its filesystem and stores the data
 - ▶ Can be to grant access to several clients based of groups or users
- The client
 - ▶ Accesses and modifies the data on the server
 - ▶ Does not store data on their own
 - ▶ Folder is included as a mounted drive

Why even use NFS

■ Pro

- ▶ Reduce storage costs
- ▶ Enables access on the same files by network for multiple parties
- ▶ Supports heterogeneous environments, as NFS is compatible with Windows, Apple, and Linux
- ▶ NFS service makes the physical location of the storage system irrelevant

■ Con

- ▶ Needs a fast network connection
- ▶ No offline access to files

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Short reminder, Terminal Editors

■ nano

- ▶ open a file with " `nano %file` "
- ▶ ctrl + s - save file
- ▶ ctrl + o - save file as
- ▶ ctrl + x - exit nano

■ vim

- ▶ open a file with " `vim %file` "
- ▶ to switch between edit and command mode press esc, in the command mode you start commands with :
- ▶ :w - writes the file
- ▶ :w !sudo tee % - writes into a file with sudo
- ▶ to exit vim
 - :q! - closes vim without saving changes
 - :wq - exits vim saving the file

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Installing an NFS server

- Installation of NFS server on CentOS/Rocky/Fedora

```
yum -y install nfs-utils
```

- Create Root NFS Directory

```
sudo mkdir /nfs
```

- Check your user and its group

1 user: `whoami`

2 group: `groups $(whoami)`

- Set the permissions for your user and his/her group

```
sudo chown $(whoami):$(id -gn $(whoami)) /nfs
```

The /etc/exports Configuration File

- This file controls
 - ▶ which file systems are exported
 - ▶ options on how they are exported
- Options
 - ▶ Blank lines are ignored
 - ▶ # starts a comment
 - ▶ \backslashes can wrap long lines\
 - ▶ for each exported item you use an individual line

Enabling access with /etc/exports

■ Access to a single client

▶ `/nfs {clientIP}(rw,sync,no_subtree_check)`

■ Access to several clients

▶ `/nfs {clientIP}(rw,sync,no_subtree_check)`

`{clientIP-2}(...)`

`{clientIP-3}(...)`

■ Access to an entire subnet

▶ `/nfs {subnetIP}/{subnetMask}(rw,sync,no_subtree_check)`

Make the NFS Share Available to Clients

- With `exportfs` we can now make the shared directory available
 - ▶ `sudo exportfs -a`
- We restart the server to reload its configuration
 - ▶ `sudo systemctl restart nfs-server`
- if you run a firewall you have to allow access
 - ▶ `sudo ufw allow`

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Setup on an NFS client

- We can install the NFS client on CentOS and Fedora using the same command as the NFS server
 - ▶ `sudo yum install nfs-utils`
- We create a local directory as a mount point for the NFS share
 - ▶ `sudo mkdir /nfs`

Mounting the NFS file system

- Now we mount the NFS share by running the mount command

- ▶ `sudo mount -t nfs {ip of NFS server}:{path on server} /nfs`

- ▶ `sudo mount -t nfs 10.254.1.234:/nfs /nfs`

The mount point now acts as the root of the nfs-share, displaying all subfolders contained in the folder on the server

- To verify the correct mounting of the NFS share run either

- ▶ `mount`

- ▶ `df -h`

Mounting the NFS file permanently

- Remote NFS directories can be mounted on startup automatically. This is defined in the file `/etc/fstab`
- We edit the `/etc/fstab`, adding a line for every file share we want to include without intermediate line breaks, similar to `etc/export` when installing the server. Parameters are separated by tabulator.
 - ▶ `nano /etc/fstab`
 - ▶ `vim /etc/fstab`

Parameters

- The last three parameters are NFS options, which we leave on default

- ▶ `{ip of NFS server}:{path on server} /nfs nfs defaults 0 0`

- examples of NFS options would be

- ▶ `timeo=n`

- How long the client waits for a response before it retries an NFS request.
Default is 60 seconds (600 * 1/10 second)

- ▶ `rsize=n`

- The maximum number of bytes in each network READ request. The largest read payload supported by the Linux NFS client is 1,048,576 bytes (one megabyte). The rsize value is a multiple of 1024 with a minimum of 4096 and maximum of 1048576 rounded down to the nearest multiple of 1024.

Try it!